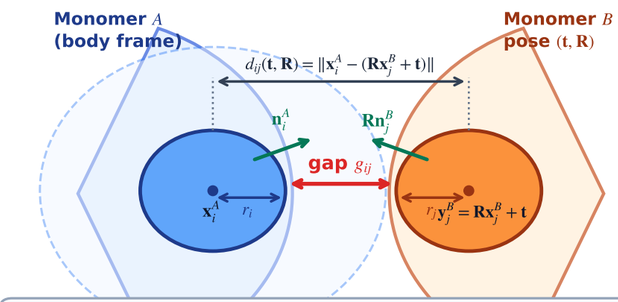
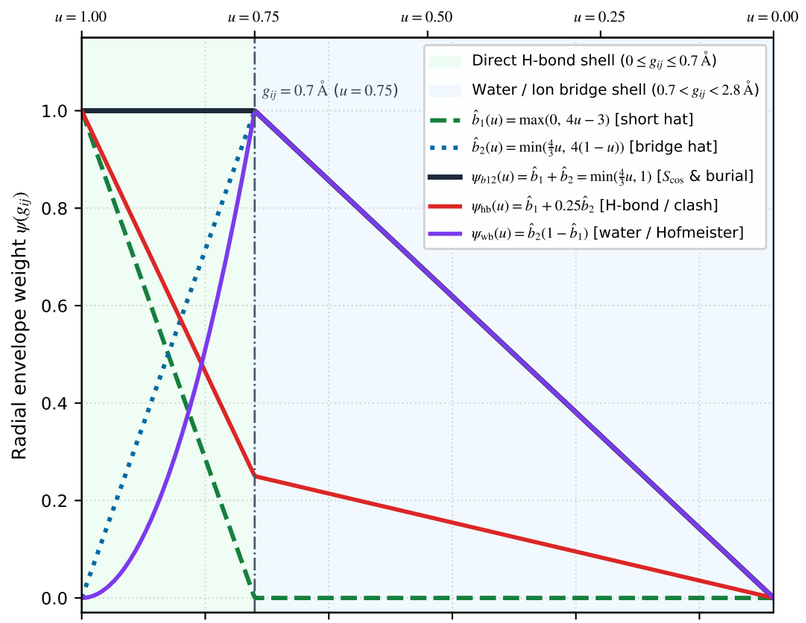
Graphical & Mathematical Definitions of Surface-Gap Envelopes (g_{ij}, u_{ij}, ψ), Complementarity (S_c vs. $S_{\cos}$), & Hofmeister Polymorph Selection

(A) Interatomic Surface Gap $g_{ij}(\mathbf{t}, \mathbf{R})$ & Proximity u_{ij}



1. **Surface Gap** ($\Delta_{\max} = 2.8$ Å): $g_{ij}(\mathbf{t}, \mathbf{R}) = d_{ij}(\mathbf{t}, \mathbf{R}) - (r_i + r_j)$
2. **Hard Sphere & Normalized Proximity:** $g_{ij} < 0 \Rightarrow E_{\mathrm{HS}} = \infty, \quad u_{ij} = \mathrm{clamp}\left(\frac{2.8 - g_{ij}}{2.8}, 0, 1\right)$
3. **Outward Surface Normal** ($\sigma = 3.5$ Å): $\mathbf{n}_i^A = -\nabla\rho_{\mathrm{self}}(\mathbf{x}_i^A)/\|\nabla\rho_{\mathrm{self}}(\mathbf{x}_i^A)\|$
4. **Solvent Exposure Factor:** $s_{\mathrm{exp},i} = \exp(-\rho_{\mathrm{self},i}/45), \quad \rho_{\mathrm{self},i} = \sum_{k \neq i} r_k^3 e^{-\|\mathbf{x}_i - \mathbf{x}_k\|^2/\sigma^2}$

(B) Radial Basis ($\hat{b}_1, \hat{b}_2$) & Envelopes $\psi(g_{ij})$



(C) Lawrence–Colman S_c vs. Multi-Point Contrastive $S_{\cos}(\mathbf{t}, \mathbf{R})$

Shared Symmetric Interface Normalization ($c_0 = 0.25$):

$$S(A, B) = \frac{1}{2}\left[\sum_{i \in A} a_i \frac{\sum_{j \in B} \psi_{b12}(g_{ij})\, a_j\, s_{ij}}{\sum_{j \in B} \psi_{b12}(g_{ij})\, a_j + c_0} + (A \leftrightarrow B)\right]$$

- **Lawrence–Colman** S_c: $s_{ij}^{\mathrm{LC}}(\mathbf{R}) = -\mathbf{n}_i^A \cdot (\mathbf{R}\mathbf{n}_j^B) \in [-1, 1] \quad (a_i = 1/N_{\mathrm{int}})$
- **Contrastive** $S_{\cos}$: $s_{ij} = \frac{1}{2}(\mathbf{u}_i^\top \mathbf{v}_j + \mathbf{u}_j^\top \mathbf{v}_i) \in [-1, 1], \quad \mathbf{u}_i, \mathbf{v}_i \in S^{D-1}\ (D = 8)$
- **3-Body Junction Gate**: $g_{\cos}(A, B) = \mathrm{clamp}(S_{\cos}(A, B)/0.18,\ 0,\ 1)$

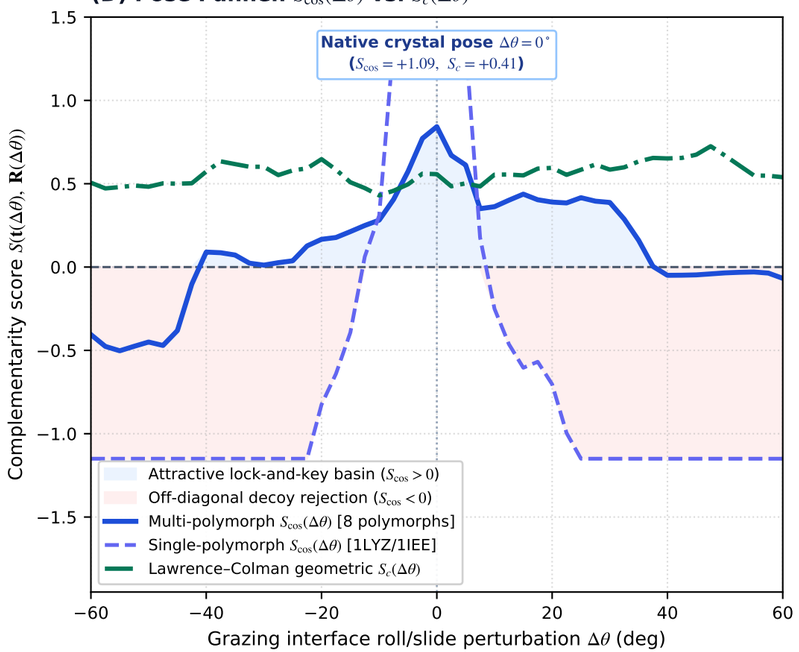
Native Lock-and-Key Pose
$(\mathbf{t}^*, \mathbf{R}^*)$: $S_{\cos} > 0$

All $\psi_{b12}(g_{ij_k}) > 0$ on matched partner pairs $s_{ij_k} \approx +0.8$

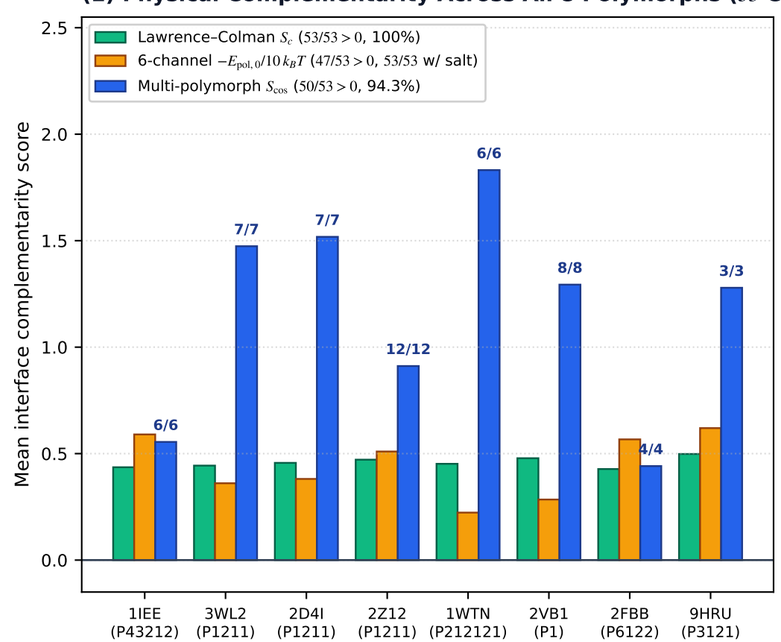
Sheared / Rotated Decoy
$(\mathbf{t}^* + \Delta\mathbf{t}, \mathbf{R}')$: $S_{\cos} < 0$

Shift/twist swaps contacts to off-diagonal pairs $s_{ij_m} \approx -0.65$

(D) Pose Funnel: $S_{\cos}(\Delta\theta)$ vs. $S_c(\Delta\theta)$



(E) Physical Complementarity Across All 8 Polymorphs (53 Contacts)



(F) Salt-Conditioned Hofmeister Stabilization $-\Delta E_{\mathrm{Hof}}(\mathbf{c})$

